LAB REPORT: EXPERIMENT 04

Data Cleaning, Preprocessing & Feature Scaling Techniques

Dataset: StudentsPerformance.csv	**Initial Shape:** 1005 rows, 8 columns
Scaling Methods: Min-Max, Standard, Robust	**Target Features:** math_score, reading_score, writing_score

1. Aim & Objective

The primary objective of this experiment is to perform data cleaning and feature transformation on student academic data. Specifically, the notebook cleans missing values, eliminates duplicate records, standardizes column syntax, and evaluates three distinct scaling methods (Min-Max Scaling, Standardization, and Robust Scaling) on continuous numerical attributes.

2. Preprocessing & Data Cleaning Steps

- **Column Name Formatting:** Converted all column headers to lowercase, stripped whitespace, and replaced spaces with underscores (e.g., 'math score' → 'math_score').
- **Imputation of Missing Values:** Identified 7 missing values in the categorical feature 'parental_level_of_education' and filled them using the column mode: `df['parental_level_of_education'].mode()[0]`
- **Deduplication:** Identified and removed 5 duplicate rows, reducing the total dataset size from 1005 to 1000 cleaned records.

3. Feature Scaling Methods Comparison

Scaler Method	Mathematical Formula	Output Range	Sensitivity to Outliers
Min-Max Scaler	X_scaled = (X - X_min) / (X_max - X_min)	[0.0, 1.0]	High (distorted by extreme min/max)
Standard Scaler	z = (X - μ) / σ	(-∞, +∞) [Mean=0, Std=1]	Moderate to High (uses Mean & Std)
Robust Scaler	X_scaled = (X - Q2) / (Q3 - Q1)	(-∞, +∞) [Median=0, IQR=1]	Low (Robust to outliers)

4. Complete Python Implementation Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler, StandardScaler, RobustScaler

# 1. Load Data & Clean Headers
df = pd.read_csv('exp04_StudentsPerformance.csv')
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

# 2. Impute Missing Values & Remove Duplicates
df["parental_level_of_education"] =
df["parental_level_of_education"].fillna(df["parental_level_of_education"].mode()[0])
df = df.drop_duplicates()

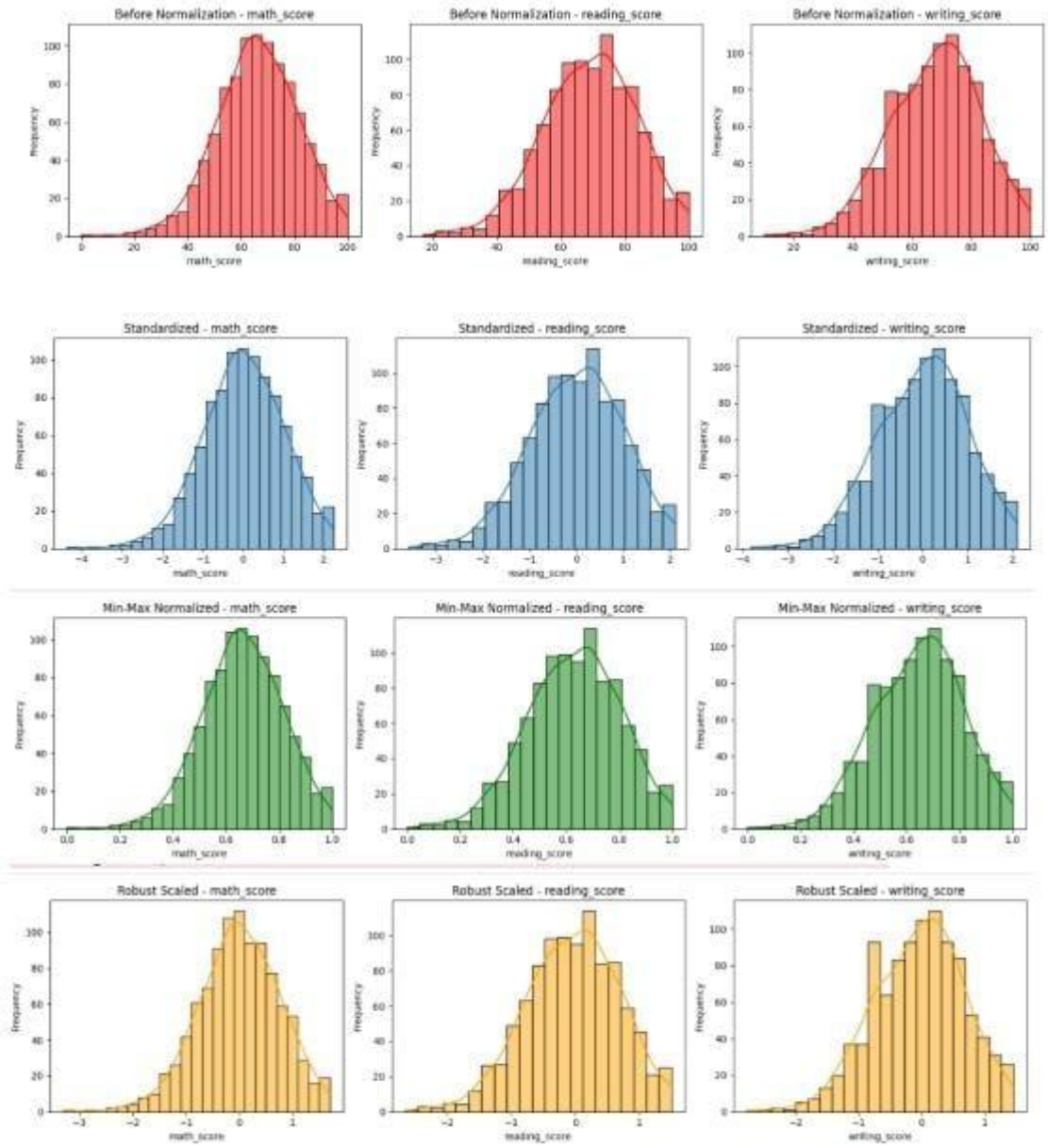
# 3. Define Numerical Features
numerical_cols = ['math_score', 'reading_score', 'writing_score']

# 4. Apply Scaling Techniques
df_minmax = df.copy()
df_minmax[numerical_cols] = MinMaxScaler().fit_transform(df[numerical_cols])

df_standard = df.copy()
df_standard[numerical_cols] = StandardScaler().fit_transform(df[numerical_cols])

df_robust = df.copy()
df_robust[numerical_cols] = RobustScaler().fit_transform(df[numerical_cols])
```

5. Output



6. Summary Result (Short Version)

★ Final Experiment Conclusions

• Data Cleaning: Imputed 7 missing values with mode and dropped 5 duplicate rows (1005 → 1000 entries).

• Feature Scaling: Successfully normalized scores to [0, 1] (Min-Max), centered them to mean 0 / std 1 (Standard), and reduced outlier impact (Robust).